

BLE Mesh based IR AC Controller

Software Design Document

Jan 23, 2025

Revision History

Version	Release Date	Prepared by	Remarks
0.2	01.04.2024	Kulasekaran	Draft Version
0.3	10.04.2024	Kulasekaran	- Removed the Heartbeat related content as temperature data has been decided to be taken as heartbeat - Added Teaching mode packet and ACK for Gwy and Node
0.4	24.04.2024	Kulasekaran	Changed the Temperature upper and lower limit keys
0.5	24.06.2024	Kulasekaran	- Added Solid Purple indication for when sending out IR command in both Gwy and Node - Added packet information regarding Teaching modes - Added Boot up LED pattern information for both Gwy and Node in the LED Table - Modified Button press logics - Added Debug Info packets for Gwy and Node - Changed Error code values - Minor changes with more clear descriptions to certain sections across the whole document - Added Detailed Error Code information - LED indication change for Teaching mode and AC Remote configuration mode.
0.6	30.07.2024	Kulasekaran	- Added ResetDevice and Logging to the Error code - Added Unsupported Remote LED indication - Removed NODE_ALREADY_PROV error code - Added AC_REMOTE_UNSUPPORTED error code - The default Heartbeat interval increased from 10s to 300s. - Added JSON_PACKET_INVALID error code
0.7	01.08.2024	Kulasekaran	- ACK responses simplified. Unnecessary fields removed. - Change to Button press logics
0.8	-	-	-
0.9	03.12.2024	Kulasekaran	Node provisioning flow modified

			• Button press functionalities modified • Error codes modified • JSON Packet modified • Teaching mode modified • LED indications modified
1.0	24.01.2025	Kulasekaran	• Added Invalid format error codes

1 Introduction

This Software Design Document outlines the technical architecture, design components, and implementation strategy for the project “IR BLE Mesh AC Controller”. It serves as a comprehensive guide to ensure the system aligns with the client's functional requirements, scalability goals, and performance expectations. The document details key design decisions, system interactions, and data flow, providing clarity for all stakeholders. Our approach prioritizes modularity, reliability, and adaptability to accommodate future enhancements. This document will act as a reference throughout development, ensuring consistency and traceability from design to deployment. Feedback is encouraged to confirm alignment with the client’s objectives and to address any concerns proactively.

1.1 Purpose of Document

The purpose of this Software Design Document (SDD) is to provide a detailed blueprint for the development of the project “IR BLE Mesh AC Controller”. It ensures a shared understanding of the system’s architecture, design principles, and implementation strategy among all stakeholders, including the client, development team, and QA team. The goal is to deliver a solution that aligns with the client’s objectives while adhering to industry best practices.

1.2 Scope of the Project

1.3 Acronyms, Abbreviations and Definitions

Acronym	Full Form
ACK	Acknowledgement
UART	Universal Asynchronous Receiver Transmitter

Abbreviation	Full Form
BLE	Bluetooth Low Energy
GWY	Gateway
IR	Infrared
JSON	Java Script Object Notation
LTE	Long Term Evolution
MQTT	Message Queuing Telemetry Transport

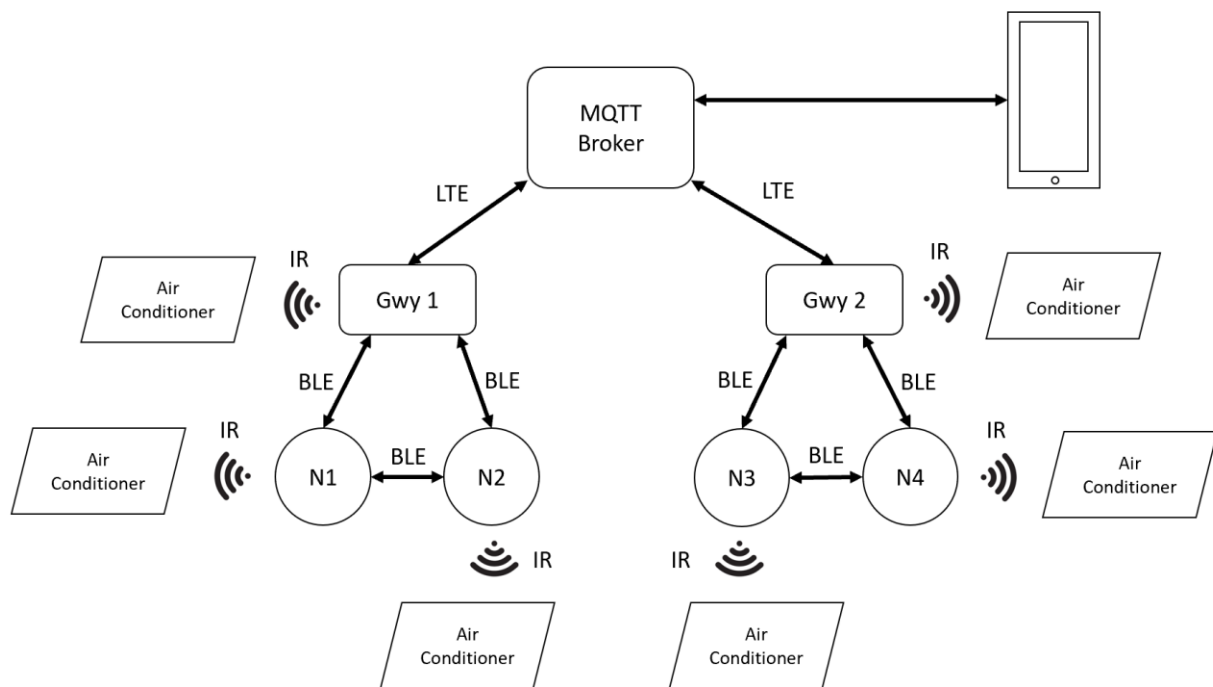
Term	Description
Registration	Adding a Gateway device
Unregistration	Removing a Gateway device
Provisioning	Adding a Node device into BLE Mesh Network
Unprovisioning	Removing a Node device from BLE Mesh Network
Configuration	Process of telling the device, which AC remote to act as

1.4 References

1.5 Overview of the Document structure

2 System Overview

The project “IR BLE Mesh AC Controller” is designed to enable seamless control and monitoring of air conditioners. The product comes in two variants: the **Gateway** and the **Node**, which work together to provide a comprehensive solution for smart AC management.



2.1 System Architecture

1. **Gateway (Master):**
 - Equipped with LTE capability for internet connectivity.
 - Connects to an MQTT broker, enabling communication through MQTT topics.
 - Acts as the central controller of the BLE mesh network.
 - Updates the system status to the cloud and processes user commands from the mobile app via MQTT.
2. **Node (Slave):**
 - Functions as a BLE mesh device communicating with the Gateway.
 - Executes commands relayed by the Gateway, such as adjusting AC settings or retrieving sensor data.
3. **Mobile App:**
 - Connects to the same MQTT broker as the Gateway.
 - Allows users to send commands and receive real-time status updates from the Gateway via MQTT topics.

2.2 Key Features

- **BLE Mesh Communication:**
 - Establishes reliable and efficient communication between Gateway and multiple Nodes within the network.
- **MQTT Integration:**
 - Facilitates seamless interaction between the Gateway and the mobile app, enabling remote control and monitoring.

2.3 Use Cases

1. **Remote Control via Mobile App:**
 - Users can adjust AC settings or query system status through MQTT-based communication with the Gateway.
2. **System Notifications:**
 - Fault detection and system status updates are pushed to the mobile app through MQTT topics for real-time alerts.

This system integrates BLE mesh communication for local functionality with MQTT-based cloud connectivity, providing a scalable and efficient solution for modern AC management.

2.4 Architecture Design

The Architecture Design section provides a comprehensive overview of the system's high-level structure, detailing the key components, their responsibilities, and how they interact with each other. This design ensures the system meets its functional and non-functional requirements, such as scalability, security, and performance.

The system is organized into multiple layers and modules, each serving a specific purpose:

1. **User Layer:**
 - **Mobile App:** The mobile app interfaces with the system, providing users with a means to control and monitor their air conditioners. The app communicates with the Gateway via MQTT to send commands and receive real-time updates.
2. **Gateway Layer:**
 - **Gateway Device:** The master unit in the system that manages communication with the Nodes via BLE Mesh and connects to the cloud via LTE for remote access. Every Gwy device is identified by its unique <GwySerNo>. This is set during flashing the Gwy device and cannot be changed later. (Eg. GWY00001)
 - **MQTT Client:** The Gateway connects to an MQTT broker, enabling bidirectional communication with the mobile app.
 - **IR Software module:** Uses a third-party IR library to decode and transmit IR signals for controlling AC units.
3. **Node Layer:**
 - **Node Device:** A BLE Mesh device that acts as a slave, responsible for controlling the AC based on commands received from the Gateway. It reports status information back to the Gateway and executes local control functions when necessary. Every Gwy device is identified by its unique <NodeSerNo>. This is set during flashing the Gwy device and cannot be changed later. (Eg. N00001).

4. Communication Layer:

- **BLE Mesh:** Provides a robust, low-power communication network between the Gateway and Node devices. BLE Mesh ensures the scalability and reliability of communication in environments with multiple devices.
- **LTE & MQTT:** The Gateway communicates with the MQTT broker over the LTE network, enabling cloud connectivity, remote status updates, and command execution.

5. Cloud Layer:

- **Cloud Platform:** Hosts the MQTT broker and processes the data sent by the Gateway. This layer is responsible for long-term storage, analytics, and any external interactions

6. Flash Memory Layer:

- **Non-Volatile Storage:** Provides the ability to store contents in flash memory to preserve them across power cycles

2.5 Key Interactions and Data Flow

- **Mobile App to MQTT Broker:** The mobile app communicates with the MQTT broker to send control commands to the Gateway.
- **Gateway to Nodes:** The Gateway manages and controls the Nodes through BLE Mesh, sending commands and receiving status updates.
- **Gateway to Cloud:** The Gateway updates the cloud platform with status data, device health information, via MQTT.
- **IR Control:** When the Gateway sends a command to control the AC, the IR Software module translates the MQTT command into the appropriate IR signals and transmits them to the AC unit.

This architecture ensures a robust, scalable, and secure system for managing air conditioning units, providing real-time control, monitoring, and seamless integration with external cloud services. The system is designed to handle a growing number of devices, ensuring reliability and performance even as the network expands.

3 Limitations & Assumptions

3.1 Limitations

- **Connectivity Range:** The BLE mesh network's effective range may be limited by environmental factors such as physical obstructions or interference from other wireless devices.
- **Internet Connectivity:** While the Gateway has LTE connectivity, the system's remote control and status updates depend on the availability of a stable internet connection. Due to the nature of MQTT and potential delays in cloud communication, there may be minor latencies in status updates or command execution, especially during high network traffic.
- **IR Library Support:** The system uses a third-party IR library for decoding IR signals and transmitting AC control IR commands. As the library is still under continuous development, not all AC models will be supported. Additionally, within the same brand, different models may use different IR protocols, meaning the model is more important than the brand when ensuring

compatibility. When talking about support, we need to consider two things: Ability to Decode IR Protocol and ability send IR Protocol.

- **Teaching Mode:** Since Teaching Mode works by capturing and storing the IR signal RAW values and replaying them back to control an AC, only a maximum of 16 commands can be stored currently. (Power off command, Temperature 18 to Temperature 32 commands (with Power ON)). Which means, when device uses teaching mode, it will be able to control only Power and Temperature.

3.2 Decodeable Protocols

Below table represents the list of protocols that are decodable, meaning recognizable by the devices (Gwy and Node). A protocol may be decodable, that doesn't mean the device can send an IR command using that protocol. In this case, teaching mode with error checking must be used for Configuration.

Brand	Protocol
Carrier	Carrier
	Carrier40
	Carrier64
	Carrier84
	Carrier128
Daikin	Daikin
	Daikin2
	Daikin64
	Daikin128
	Daikin152
	Daikin160
	Daikin176
	Daikin200
	Daikin216
	Daikin280
	Daikin312
Haier	Haier
	HaierYRW02
	Haier160
	Haier176
Hitachi	Hitachi1
	Hitachi2
	Hitachi3
	Hitachi264
	Hitachi296
	Hitachi344

	Hitachi424
LG	LG
	LG2
Mitsubishi	Mitsubishi
	Mitubishi1
	MitsubishiAC
	Mitubishi112
	Mitsubishi136
	MitsubishiHeavy
Samsung	Samsung
	SamsungAC
	Samsung36
Toshiba	Toshiba56
	Toshiba72
	Toshiba80
Voltas	Voltas

3.3 Sendable Protocols

Below table represents the list of protocols that the devices could send an IR command in. It also provides the information as to what functionalities are supported by the library in each of these protocols.

Protocol	Power	Temp	Fan	Mode	SwingH	SwingV	OnTimer	OffTimer
Carrier64	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Daikin2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Daikin64	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Daikin128	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Daikin152	Yes	Yes	Yes	Yes	No	Yes	No	No
Daikin160	Yes	Yes	Yes	Yes	No	Yes	No	No
Daikin176	Yes	Yes	Yes	Yes	Yes	No	No	No
Daikin200	Yes	Yes	Yes	Yes	Yes	No	No	No
Daikin216	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Daikin280	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Haier	No	Yes	Yes	Yes	No	Yes	No	No
Haier160	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Haier176	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hitachi	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Hitachi1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hitachi296	Yes	Yes	Yes	Yes	No	No	No	No

Hitachi424	Yes	Yes	Yes	Yes	No	Yes	No	No
LG	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Mitsubishi	Yes	Yes	Yes	Yes	No	No	No	No
Mitsubishi88	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Mitsubishi112	Yes	Yes	Yes	Yes	No	Yes	No	No
Mitsubishi136	Yes	Yes	Yes	Yes	No	Yes	No	No
Mitsubishi152	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Samsung	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Toshiba	Yes	Yes	Yes	Yes	No	Yes	No	No
Voltas	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

3.4 Assumptions

- **Network Stability:** It is assumed that the Gateway will always have access to a reliable LTE network for remote communication and MQTT-based control.
- **IR noise free environment:** During initial setup and during the process of Teaching, the devices must be in an IR noise free environment.

4 Physical Button Functionalities

Press type	Gateway	Node
Single press	None	None
Double press	Initiate Power Cycle sequence	Initiate Power Cycle sequence
Triple press	Unregisters device	Unprovisions device
Long press (1-3s)	Enter/Exit Teaching mode	Enter/Exit Teaching mode
Long press (>3s)	Factory Reset Device	Factory Reset Device

5 LED Indications

Indication	Gateway	Node
Solid Red	Device not connected to MQTT	Device is Unprovisioned and non-connectable
Toggling Red and Blue	Device is Unregistered	Device in Unprovisioned and connectable
Solid Blue	Device is Unconfigured	Device is Unconfigured
Solid Green	Fully operational	Fully operational
Blinking Blue	Device in Teaching mode	Device in Teaching mode
Blinking Yellow	Device in power cycle sequence	Device in power cycle sequence
Blinking Purple	Sending out IR signal	Sending out IR signal
Blinking Red	Invalid event/command detected	Invalid event/command detected
Blinking Green	Device detected IR Signal	Device detected IR Signal

6 JSON Packets

Packet ID	Packet Name
0	Gwy Registration Packet
1	Gwy AC Remote Configuration Packet
2	Gwy Unregistration Packet
3	Gwy AC Control Packet
4	Gwy Manual AC Control ACK
5	Gwy AC Remote Reconfiguration Packet
6	Gwy Heartbeat ACK
7	Gwy Heartbeat Publish Configuration Packet
8	Gwy Teaching Mode Packet
9	Gwy Teaching Mode Command Selection Packet
10	Gwy Debug Info Packet
12	Gwy General ACK
100	Node Provisioning Packet
101	Node AC Remote Configuration ACK
102	Node Unprovisioning Packet
103	Node AC Control Packet
104	Node Manual AC Control ACK
105	Node AC Remote Reconfiguration Packet
106	Node Heartbeat ACK
107	Node Heartbeat Publish Configuration Packet
108	Node Teaching Mode Packet
109	Node Teaching Mode Command Selection Packet
110	Node Debug Info Packet
112	Node General ACK

For more information about these packets, what parameters are required, the data type of values and the value range, please refer to this [link](#). This will contain examples of these packets as well.

7 Error Codes

Error Code	Error Code Name
-1	FAILURE
0	SUCCESS
1	JSON_PACKET_INVALID
2	MISSING_PACKET_ID
3	PACKET_ID_EXCEEDING_RANGE
4	MISSING_MSG_SEQ_NO

5	MSG_SEQ_NO_EXCEEDING_RANGE
6	MISSING_LOCATION
7	LOCATION_EXCEEDING_RANGE
8	NODE_COMM_TIMEOUT
9	GWY_NOT_REG
10	GWY_ALREADY_REG
11	MISSING_MACID
12	INVALID_MACID
13	MISSING_ELEMENT_ADDR
14	ELEMENT_ADDR_EXCEEDING_RANGE
15	DEVICE_NOT_CONFIGURED_WITH_AC_REMOTE
16	MISSING_POWER
17	POWER_EXCEEDING_RANGE
18	MISSING_MODE
19	MODE_EXCEEDING_RANGE
20	MISSING_FAN_SPEED
21	FAN_SPEED_EXCEEDING_RANGE
22	MISSING_TEMPERATURE
23	TEMPERATURE_EXCEEDING_RANGE
24	MISSING_SWINGH
25	SWINGH_EXCEEDING_RANGE
26	MISSING_SWINGV
27	SWINGV_EXCEEDING_RANGE
28	MISSING_ONTIMER
29	ONTIMER_EXCEEDING_RANGE
30	MISSING_OFFTIMER
31	OFFTIMER_EXCEEDING_RANGE
32	MISSING_LOCKING
33	LOCKING_EXCEEDING_RANGE
34	MISSING_TEMPERATURE_UPPER_LIMIT
35	TEMPERATURE_UPPER_LIMIT_EXCEEDING_RANGE
36	MISSING_TEMPERATURE_LOWER_LIMIT
37	TEMPERATURE_LOWER_LIMIT_EXCEEDING_RANGE
38	INVALID_TEMPERATURE_LOCKING_LIMITS
39	MISSING_PUBLISH_PERIOD
40	PUBLISH_PERIOD_EXCEEDING_RANGE
41	MISSING_RESET_DEVICE
42	RESET_DEVICE_EXCEEDING_RANGE
43	MISSING_LOGGING
44	LOGGING_EXCEEDING_RANGE

45	IR_PROTOCOL_DECODEABLE_ONLY
46	IR_PROTOCOL_FULLY_UNSUPPORTED
47	MISSING_ERROR_CHECK
48	ERROR_CHECK_EXCEEDING_RANGE
49	MISSING_TEACHING_START
50	TEACHING_START_EXCEEDING_RANGE
51	MISSING_STARTING_TEMPERATURE
52	STARTING_TEMPERATURE_EXCEEDING_RANGE
53	MISSING_ENDING_TEMPERATURE
54	ENDING_TEMPERATURE_EXCEEDING_RANGE
55	STARTING_TEMPERATURE_LESS_THAN_ENDING_TEMPERATURE
56	ENTERED_TEACHING_MODE
57	EXITED_TEACHING_MODE
58	DEVICE_ALREADY_IN_TEACHING_MODE
59	DEVICE_NOT_IN_TEACHING_MODE
60	POWER_NOT_AVAILABLE_IN_IR_SIGNAL_DECODED_STRING
61	MODE_NOT_AVAILABLE_IN_IR_SIGNAL_DECODED_STRING
62	FANSPEED_NOT_AVAILABLE_IN_IR_SIGNAL_DECODED_STRING
63	TEMPERATURE_NOT_AVAILABLE_IN_IR_SIGNAL_DECODED_STRING
64	IR_CMD_NOT_AVAILABLE_IN_FLASH
65	NODE_NOT_FOUND_IN_PROVISIONER_DATABASE
66	NODE_NOT_PROVISIONED
67	IR_TASK_CREATION_FAILED (QMAX 's Internal Use only)
68	ENQUEUEING_INTO_COMMAND_QUEUE_FAILED
69	DEVICE_DATA_ERASURE_SUCCESSFUL
70	DEVICE_DATA_ERASURE_FAILED
71	IR_PARTITION_INIT_FAILED (QMAX 's Internal Use only)
72	POWER_INVALID_FORMAT
73	TEMPERATURE_INVALID_FORMAT
74	FAN_SPEED_INVALID_FORMAT
75	MODE_INVALID_FORMAT
76	SWINGH_INVALID_FORMAT
77	SWINGV_INVALID_FORMAT
78	LOCKING_INVALID_FORMAT
79	ONTIMER_INVALID_FORMAT
80	OFFTIMER_INVALID_FORMAT
81	TEMPERATURE_UPPER_LIMIT_INVALID_FORMAT
82	TEMPERATURE_LOWER_LIMIT_INVALID_FORMAT
83	PACKETID_INVALID_FORMAT
84	MSG_SEQ_NO_INVALID_FORMAT

85	ELEMENT_ADDR_INVALID_FORMAT
86	LOCATION_INVALID_FORMAT
87	PUBLISH_PERIOD_INVALID_FORMAT
88	MACID_INVALID_FORMAT
89	TEACHING_START_INVALID_FORMAT
90	STARTING_TEMPERATURE_INVALID_FORMAT
91	ENDING_TEMPERATURE_INVALID_FORMAT
92	RESTART_INVALID_FORMAT
93	RESET_INVALID_FORMAT

Most of the above-mentioned error codes are self-explanatory as to what they mean and when they will occur. Some of the ones are explained below:

- Error codes containing “MISSING” mean, a JSON packet was received by the Gwy without a required parameter in it.
- Error codes containing “EXCEEDING_RANGE” mean, a JSON packet was received by the Gwy with one of the parameters containing a value out of range.

8 MQTT Topics

Gwy uses the following topics for communication with the MQTT Broker.

Topic	Usage
<GwySerNo>/command	Gwy receives commands from this topic
<GwySerNo>/message	Gwy sends out ACKs to this topic
<GwySerNo>/alive	Gwy sends a message to this topic once every minute to keep the MQTT connection alive
GWYS/will	All Gwy devices will make a will configuration to this topic. Meaning, when Gwy loses its MQTT connection, the MQTT broker, on behalf of the Gwy will send out a message “<GwySerNo> disconnected Unexpectedly” to the clients that subscribed to this topic

9 Teaching Mode

Teaching Mode is a method in which IR commands fired from the AC remote is captured and stored by the devices (Gwy and Node) and then replayed back to control the AC. Teaching mode is intended to be used when the IR protocol used by the AC Remote is not in the list of [sendable protocols](#).

Teaching Mode comes off as two variants.

1. Teaching Mode with Error Check
2. Teaching Mode without Error Check

9.1 Teaching Mode with Error Check

If the IR Protocol used by the AC remote is not in the list of [sendable protocols](#), but is in the list of [decodable protocols](#), then Teaching Mode with Error Check will be used by the device. What this means is that the device will have the ability to know whether the correct IR command as per the [Teaching mode command sequence](#) was received by the device. For example, the first IR command to be received by the device is “Power OFF” command. But if the user accidentally sends a different command, then the device will not store it and will send an ack with appropriate error code.

9.2 Teaching Mode without Error Check

If the IR Protocol used by the AC remote is neither sendable nor decodable, then Teaching Mode without Error check will be used by the device. In this case, to make the process reliable and robust, the user will have to choose the command that he’s going to send from the remote and then send the command. For example, the user can choose “Power OFF” command and then fire the Power OFF command from the AC remote. By this way, the device will store the command in the right slot. Even if any error occurs, the user can re-choose the slot and overwrite the wrongly stored command with the right one. The device will automatically exit the teaching mode, once the necessary number of commands have been received.

9.3 Teaching Mode command sequence

Command Number	Power	Temperature
1	Off	Anything
2	On	18
3	On	19
4	On	20
5	On	21
6	On	22
7	On	23
8	On	24
9	On	25
10	On	26
11	On	27
12	On	28
13	On	29
14	On	30
15	On	31
16	On	32

10 Setup Instructions

This section contains step by step instructions on how to set up devices according to different configurations.

10.1 One Gwy – Normal Mode

1. Power ON Gateway
2. Register Gateway
3. If Gateway LED is solid Blue, Move on to next step.
4. Configure the Gateway with AC remote by pointing the AC remote towards the Gateway and pressing any button (preferably the Power ON/OFF button)
5. If Gateway LED is solid Green, Gateway is now ready to control the AC.

10.2 One Gwy – Teaching Mode with Error Check enabled

1. Power ON Gateway
2. Register Gateway
3. If Gateway LED is solid Blue, Move on to next step.
4. Configure the Gateway with AC remote by pointing the AC remote towards the Gateway and pressing any button (preferably the Power ON/OFF button)
5. The Gateway will check if the IR Protocol used by the AC remote is in the list of the [Sendable Protocols](#). In this case, assume it's not in the list. Then, the next step is to proceed with Teaching Mode.
6. Send Gateway Teaching Mode packet.
7. Gateway Enters Teaching Mode with Error Check enabled.
8. Fire IR commands in the right sequence. Pay attention to LED indication and Ack received by the MQTT Broker to know if the command was successful
9. Once the necessary number of commands gets received, the Gateway will automatically quit teaching mode and get configured.
10. If the Gateway is solid Green, Gateway is now ready to control the AC.